MediTrust

Sprint_3_Review Group7

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1. Sprint 3 Objective

The main objective of Sprint 3 was to enhance the MediTrust system by improving the user interface, implementing role-based access control, restructuring the frontend into a modular architecture, and integrating explainable AI features into the prediction results.

In Sprint 2, we successfully integrated the machine learning model and built a complete prediction pipeline. However, the system lacked structured user roles, visual clarity in outputs, and interpretability of predictions.

During Sprint 3, we focused on improving usability and making the system more realistic by introducing login-based access, separate dashboards for different users, better visualization of risk, and adding explainability to understand how predictions are generated.

By the end of this sprint, the system supports:

- Secure login and role-based access
- Improved UI with better navigation
- Clear risk visualization
- Explainable AI insights for predictions
- Modular frontend structure for scalability

2. What We Completed in This Sprint

During Sprint 3, we successfully enhanced both the frontend and system workflow to make the application more structured and user-friendly.

First, we implemented a complete authentication system including login and signup functionality with role-based access. This allows different types of users such as doctors, nurses, and admins to interact with the system based on their responsibilities.

Next, we developed separate dashboards for each role. The doctor dashboard focuses on prediction results and decision support, while the nurse dashboard is designed for entering patient clinical data. This separation improves usability and reflects real-world hospital workflows.

We also redesigned the user interface to make it more professional and consistent. The layout was improved, unnecessary sections were removed, and a healthcare-oriented theme was applied to enhance the user experience.

Another important improvement was restructuring the frontend into a modular architecture. Instead of keeping all logic in a single file, the code was divided into modules such as authentication, prediction handling, API services, and utility functions. This makes the system easier to maintain and extend.

In addition, we enhanced the prediction output by displaying risk probability along with categorized risk levels such as low, medium, and high. A visual progress bar with color mapping was added to clearly represent risk severity.

Finally, we integrated explainable AI features using SHAP-like outputs to show how different input features contribute to the prediction. This makes the system more transparent and useful for clinical decision-making.

3. Demonstration of Functionality

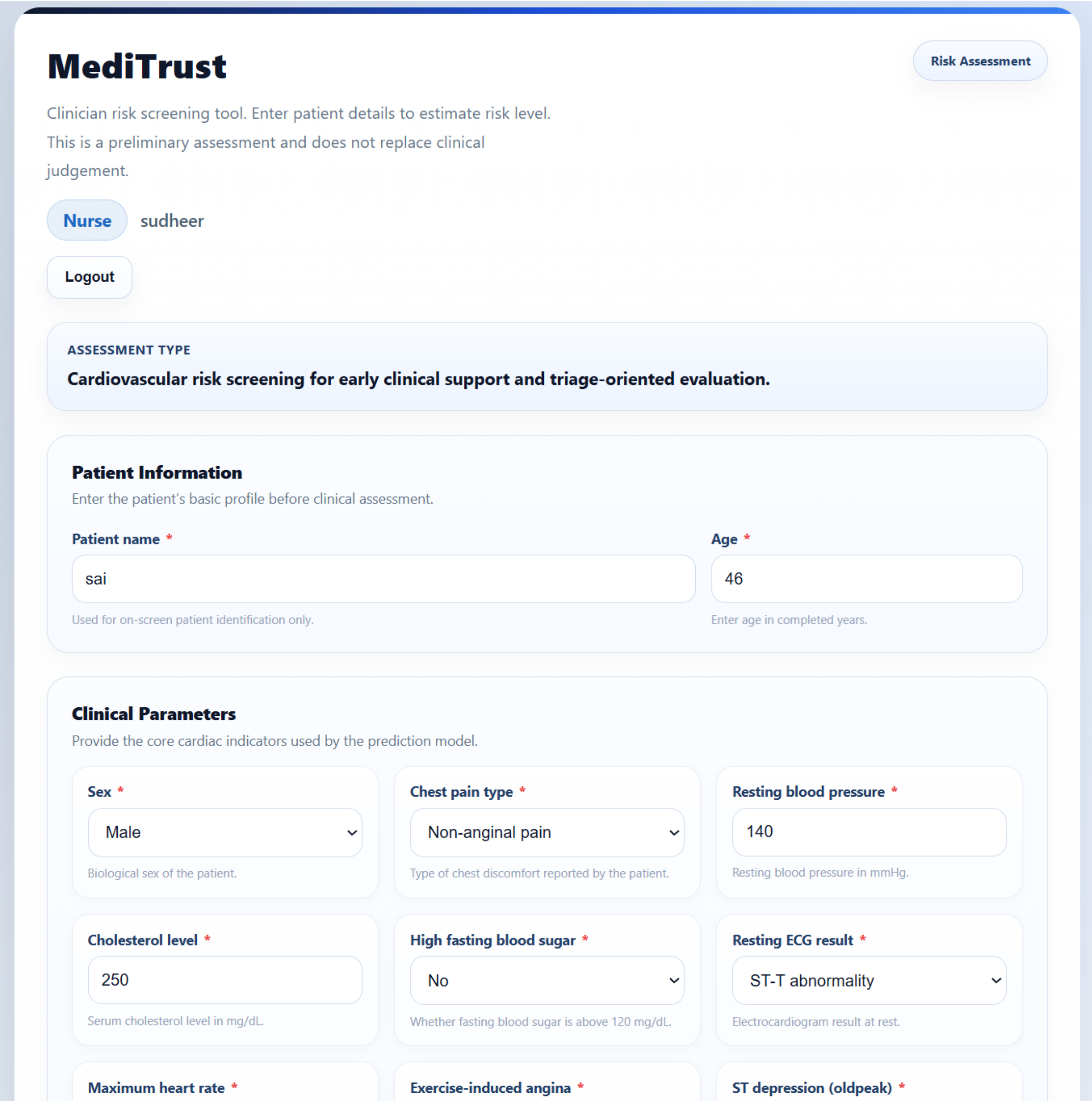
The following workflow was tested successfully:

1. User logs into the system using role-based authentication
2. User accesses their respective dashboard (Doctor/Nurse)
3. Patient clinical parameters are entered through the form
4. The frontend sends the data to the backend prediction API
5. The machine learning model processes the input data
6. The system returns the predicted cardiovascular risk level
7. The result is displayed with visualization and explanation

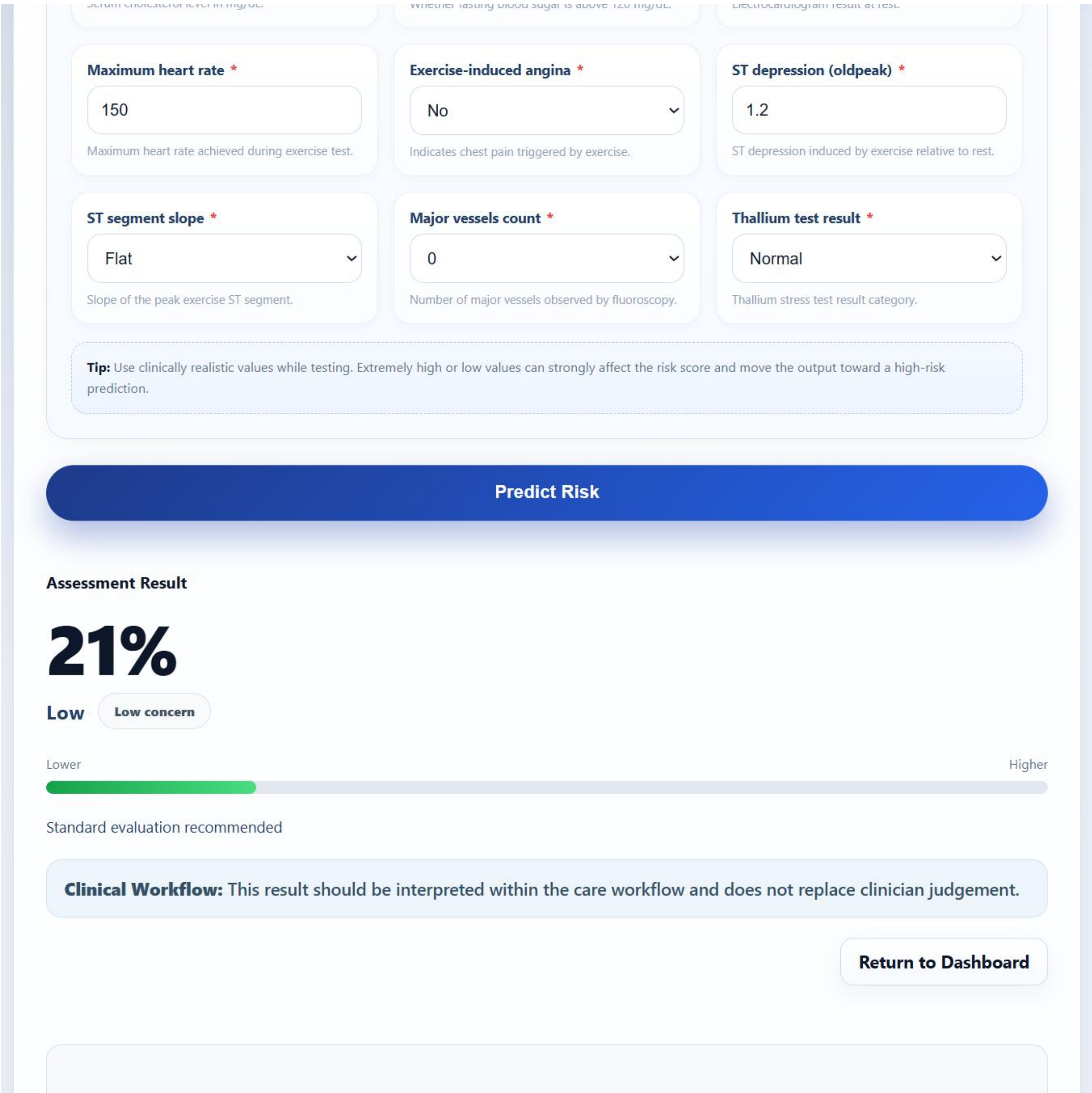
The prediction output includes:

- Risk Probability (e.g., 0.78)
- Risk Level (Low / Medium / High)
- Recommendation message
- Visual progress bar indicating risk
- Feature contribution insights (Explainability)

Example output:



From the above screenshot, it shows the cardiovascular risk assessment form where the user (Nurse role) enters patient details such as age, sex, blood pressure, cholesterol, ECG results, and other clinical parameters. The interface is structured and user-friendly, ensuring proper data entry before prediction.



From the above screenshot it displays the predicted cardiovascular risk result. The system shows the risk probability (21%) along with the risk level (Low). A progress bar is used to visually represent the risk severity, and a clinical recommendation is provided to guide further evaluation

This section provides a detailed explanation of how each clinical feature influenced the prediction. It highlights which factors increased the risk (e.g., ST slope, blood pressure) and which reduced it (e.g., chest pain type, vessels), making the model output interpretable for clinical use.

This screen shows graphical explainability using contribution bars and a SHAP waterfall view. It illustrates how the prediction moves from a base value to the final probability by adding or subtracting the impact of each feature, providing a clear understanding of the model's decision process.

(Screenshots of prediction interface, API response, GitHub commits, and project board are attached.)

4. Project Management Tracking

We continued using the GitHub Project Board to track tasks and development progress.

Project Board Link: *https://github.com/users/RavindraSSK/projects/7*

Tasks moved through the following columns:

• Backlog
• Ready
• In Progress
• In Review
• Done

All Sprint 3 tasks were completed and moved to the Done column. Additional improvements such as UI refinements and explainability integration were also tracked and completed within this sprint.

5. Branching Strategy

During Sprint 3, we followed a structured branching strategy to manage development and integration.

The main branches used were:

- **main** as stable branch
- **sprint3** as Sprint 3 integration branch

All development work was implemented in feature branches and then merged into the **sprint3** branch before final integration into **main**.

Feature Branches Used in Sprint 3

- feature/ui-ux-enhancement-clean
- feature/ui-ux-enhancement
- 47-enhance-login-system-with-standard-features

These branches were used for UI/UX improvements and authentication features.

Merge Flow

- Feature branches → merged into **sprint3**
- sprint3 → merged into **main**

Some additional branches such as feature/backend-finalization, feature/integration-testing were created during early planning and experimentation but were not actively used in the final Sprint 3 integration.

6. Challenges Faced

During Sprint 3, we encountered several challenges while improving the system and integrating new features.

One of the main challenges was maintaining consistency in the user interface after redesigning multiple components. Some pages initially had different styles and layouts, which required additional effort to standardize the design across the application.

Another issue was related to risk visualization. Initially, the risk output did not properly reflect different levels, and the color mapping was not accurate. This was resolved by implementing threshold-based logic to correctly display low, medium, and high risk using appropriate colors.

We also faced challenges while integrating explainable AI outputs into the frontend. Converting SHAP values into meaningful textual explanations and visual components required careful mapping between backend data and UI elements.

Additionally, restructuring the frontend into a modular architecture was slightly complex, as existing code had to be reorganized without breaking functionality. This required proper separation of concerns and testing after each change.

Finally, during development, there were minor integration issues between frontend and backend APIs. These were resolved through testing and debugging to ensure smooth end-to-end workflow.

After addressing these challenges, the system became more stable, user-friendly, and easier to maintain.

Note: Respective screenshots are uploaded in screenshot folder